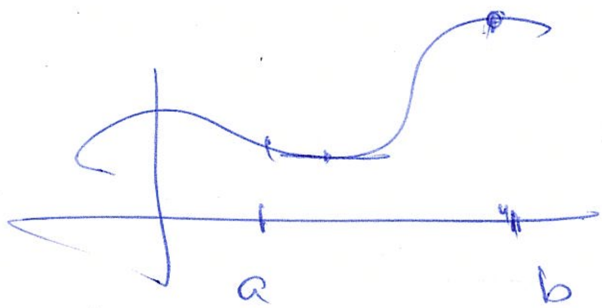
MATH3611/5705: Compactness

A continuous real valued function on a closed and bounded interval $[a, b]$ achieves a maximum. Both "closed" and "bounded" are necessary!

$f : [a,b] \to \mathbb{R}$ $\quad f \in C[a,b]$



$\exists\, x_{max} \in [a,b] : \quad f(x_{max}) \geq f(x) \ \forall x \in [a,b]$

$\exists\, x_{min} \in [a,b] : \quad f(x_{min}) \leq f(x) \ \forall x \in [a,b]$

Definition

A topological space (X, τ) is *compact* if for every $\{V_i\}_{i \in I} \subseteq \tau$ such that

$$X = \bigcup_{i \in I} V_i,$$

open cover
open cover

there is a finite subset $\{i_1, \ldots, i_n\} \subseteq I$ such that

$$X = \bigcup_{k=1}^{n} V_{i_k}.$$

finite subcover

A subset $Y \subseteq X$ is compact iff it is compact in the subspace topology.

In words: "Every open cover has a **finite** subcover".

Exercise:

1. A coarse topological space is compact.
2. A discrete topological space is compact iff it is finite.

① $(X, \tau_{coarse} = \{\emptyset, X\})$

the only open cover $X \subseteq \bigcup X$

② $(X, \tau_{discrete})$ $\tau_{discrete} = P(X)$ power set

choose one-element set
open cover

$$X = \bigcup_{x \in X} \{x\} \quad \underline{\text{open cover}}$$

If X compact $\exists$ $x_1, x_2, \ldots, x_n \in X$

$$\underline{X} = \bigcup_{i=1}^{n} \underline{\{x_i\}} \quad \text{finite subcover}$$

$\Rightarrow$ $\underline{X \text{ finite}}$

compact$_1$

Compactness in terms of closed sets: A topological space (X, τ) is compact if every collection $\{V_i\}_{i \in I}$ of closed subsets of X such that *all finite intersections $V_{i_1} \cap \ldots \cap V_{i_n}$ are nonempty* has nonempty intersection.

the core idea is de Morgan laws

compact $\overset{\Rightarrow}{\underset{\Leftarrow}{\Longleftrightarrow}}$ compact$_1$

Assume compact.

Fix $\{V_i\}_{i \in I}$ family of closed sets s.t.

$\bigcap_{k=1}^{n} V_{i_k} \neq \emptyset$ for any set $\{i_1, i_2 \ldots, i_n\}$

must show $\bigcap_{i \in I} V_i \neq \emptyset$

$U_i = X \setminus V_i$ open is it true that

$X \neq \bigcup_{i \in I} U_i \overset{\text{deML}}{\Longleftrightarrow} \emptyset = \bigcap_{i \in I} V_i$

$\Updownarrow$ by compactness

$\exists\ i_1, i_2 \ldots, i_n$: d'ML

$X = \bigcup_{k=1}^{n} U_{i_k} \Longleftrightarrow \emptyset = \bigcap_{k=1}^{n} V_{i_k}$

Question: What does is it mean for a subset $Y \subseteq X$ to be compact?

Y - compact $\Leftrightarrow$ (Y, τ_Y) compact

$\forall\ Y = \bigcup_{i \in I} U_i \quad U_i \in \tau_Y$

$\exists$ finite subcover i.e. $\exists\ i_1, i_2, \ldots, i_n$:

$Y = \bigcup_{k=1}^{n} U_{i_k}$

$\boxed{U_i \in \tau_Y \Leftrightarrow U_i = Y \cap V_i \quad V_i \in \tau_X}$

$\forall\ Y \subseteq \bigcup_{i \in I} V_i \quad V_i \in \tau_X$

$\exists\ i_1, i_2, \ldots, i_n$: $Y \subseteq \bigcup_{k=1}^{n} V_{i_k}$

Theorem

The interval $[0,1]$ *is compact.*

Proof. $S = \{x \in [0,1] : [0,x] \text{ is compact}\}$

$S \subseteq [0,1]$

(1) $S \neq \emptyset \Leftarrow 0 \in S \Leftrightarrow [0,0] = \{0\}$ compact.

$s_0 = \sup S$ def of sup.

(2) $s_0 \in S$? $\forall \varepsilon > 0 \ \exists x_\varepsilon \in S: \ s_0 - \varepsilon < x_\varepsilon \leq s_0$

since $x_\varepsilon \in S \Rightarrow [0, x_\varepsilon]$ – compact

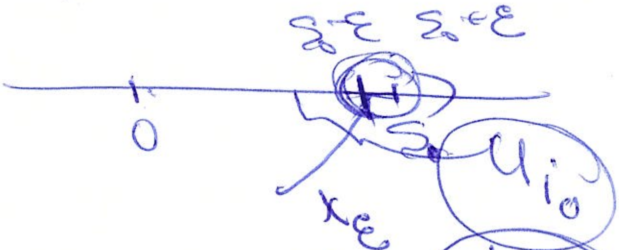
$s_0 \in S \Leftrightarrow [0, s_0]$ – compact?

$[0, s_0] \subseteq \bigcup_{i \in I} U_i$ – open cover

$\exists i_0 : s_0 \in U_{i_0} \xrightarrow{\text{open}} \exists \varepsilon : B(s_0, \varepsilon) \subseteq U_{i_0}$

by sup def $\exists$

$x_\varepsilon \in S: \ s_0 - \varepsilon < x_\varepsilon \leq s_0$



$[0, x_\varepsilon] \subseteq \bigcup_{i \in I} U_i$ open cover

compact

$\exists i_1, i_2, \ldots, i_n : [0, x_\varepsilon] \subseteq \bigcup_{k=1}^{n} U_{i_k}$

$[0, s_0] \subseteq \left[\bigcup_{k=1}^{n} U_{i_k} \right] \cup U_{i_0}$

step 3

$s_0 = 1$?

by contradiction

$s_0 < 1$

$\Downarrow$

$\exists s_0 < x_0 \leq 1$

$x_0 \in S$

Theorem

Let (X, τ_X) and (Y, τ_Y) be compact topological spaces. Then $X \times Y$ is compact (in the product topology).

$[0,1] \times [0,1]$ – square in $\mathbb{R}^2$

$[0,1]^3$ – cube in $\mathbb{R}^3$

Proof. $X \times Y = \bigcup_{i \in I} U_i \times V_i$

$\overset{?}{\Rightarrow} \exists\, i_1, i_2, \dots, i_n \quad X \times Y = \bigcup_{k=1}^{n} U_{i_k} \times V_{i_k}$

$x \in X \overset{?}{=} \bigcup_{i \in I} U_i$ true

$Y = \bigcup_{i \in I} Y_i$ exercise.